TEXORA · LECTURE SHEET

The Blow Room

The first machine room of spinning: it opens compressed bale tufts to a fine, clean flock, blends the mixing together, removes most of the trash the cotton brought from the field, and delivers a uniform lap the card can work on.

TE-2101 · SPINNING TECHNOLOGY-I · TOPIC 2 · LEVEL 2 TERM 1

1. Objects of the blow room

The blow room is a **line** of connected opening, cleaning and conveying units ending at the scutcher, which forms the lap. Its four objects must be achieved together:

- **Opening** — reduce dense bale tufts progressively to a fine flock so trash particles are exposed and can fall free. Opening must be *gradual*: violent opening of large tufts breaks fibres and creates neps, and trash locked inside a large tuft cannot be removed at all.
- **Cleaning** — remove trash, leaf, seed fragments and dust; typically 4–8 % of bale weight leaves as blow-room waste for ordinary cottons. Cleaning works best when tufts are smallest and trash is given a separation opportunity (grid bars, perforated surfaces, air separation, gravity drops).
- **Mixing and blending** — combine the laydown so intimately that every lap contains every stock in the formulated proportions. Pluckers, blending hoppers and mixing chambers do most of the mill's mixing.
- **Uniform lap formation** — deliver a lap (or, in chute-feed lines, a continuous mat) of constant weight per metre, even across its width, firm enough to unroll at the card without stretching. Lap weight variation passes directly into sliver and yarn count variation.

A supporting object is **dust removal** by suction and filtration at each machine — modern lines are enclosed and under suction for fibre quality and for health.

2. A typical blow-room line

Fibre travels between machines in ducts, carried by air from fans and condensers, so the line works continuously. A representative line:

1. **Bale plucker / bale breaker.** A travelling head or spiked lattice takes small tufts from each laydown bale in turn — gentle first opening, and already a mixing action. Older lines used a bale breaker with spiked lattices and an evener roller for hand-fed bales.

2. **Step cleaner / pre-cleaner.** Inclined spiked lattices or beater-and-grid units toss the tufts over grid bars in steps; loosened coarse trash falls through the grids. Heavy trash is removed early, before later beaters break it into harder-to-remove fragments.
3. **Hopper feeder / blending hopper.** A reservoir hopper whose lattice and evener rollers meter fibre forward at a controlled rate, smoothing plucker surges; in blending lines, separate hoppers dose different stocks by weight or rate.
4. **Fine opener** (e.g. porcupine opener). A fast beater over grid bars opens tufts finer and ejects more trash. Trashy cottons get more cleaning points; clean long-staple cottons get fewer, gentler ones.
5. **Condenser.** A perforated rotating drum with internal suction draws airborne fibre onto its surface as a sheet, separates fibre from conveying air, and drops the sheet onward; dust passes through the perforations to filtration.
6. **Scutcher.** The last machine: final beating and cleaning, pedal-regulated feed, and lap formation (Section 3). With two scutchers, the first (*breaker*) does coarser work and its lap feeds the second (*finisher*), which makes the card lap.

Two principles govern every line: **gentle first, intensive later** — opening intensity rises as tuft size falls; and **cleaning needs a drop point** — a beater without grids, cage or separation chamber beneath it merely moves trash around.

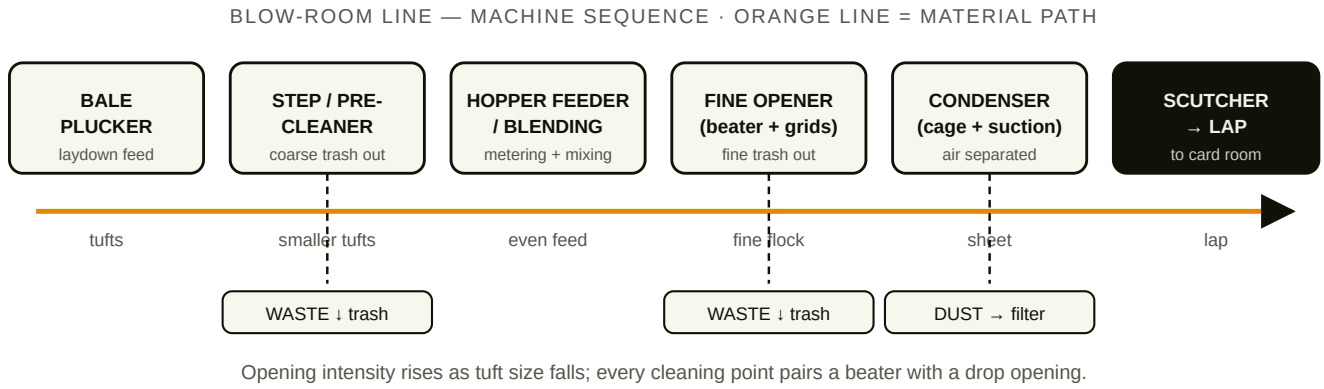


Figure 1 — A typical blow-room line in sequence. Material (orange) is air-conveyed between machines; waste drops at the cleaners and dust is drawn off at the condenser.

3. The scutcher in detail

3.1 Feed and regulation

Fibre from the condenser falls onto a moving **feed lattice**, which carries it under **feed rollers** working against a feed plate and spring-loaded **pedals**. The pedal system is the evenness control: a thicker sheet depresses its pedal further, and through the regulating motion (cone drums in classic machines, electronic sensing in modern ones) the feed roller is *slowed*; a thinner sheet lets feed speed up. Thick

and thin sections are thus compensated within the lap — a short-term regulation that cannot correct a wrongly set average weight.

3.2 Beater and grid bars

The gripped fringe is struck by the **beater** as it leaves the feed nip. **Kirschner beater** — three bladed arms carrying pins or serrated edges; a combing-striking action, relatively gentle, the standard finisher beater for better cottons. **Porcupine beater** — a drum covered with projecting teeth; aggressive tearing opening, used earlier in the line or for coarse, trashy cottons. Beneath the beater, adjustable **grid bars** form a curved grid: struck trash and seed fly out between the bars into the waste chamber while the lighter fibre is carried onward in the air stream.

3.3 Cages and lap formation

Two perforated suction drums, the **cages**, condense the fibre into an even sheet and extract dust to the filter. The sheet passes between compressing **lap (calender) rollers** onto a lap rod rotated by the winding rollers, building a firm lap. At the set length a measuring / knock-off motion stops the feed for doffing. A finished lap is typically about 10–18 kg for roughly 35–45 m, at 350–500 g/m according to the card it feeds.

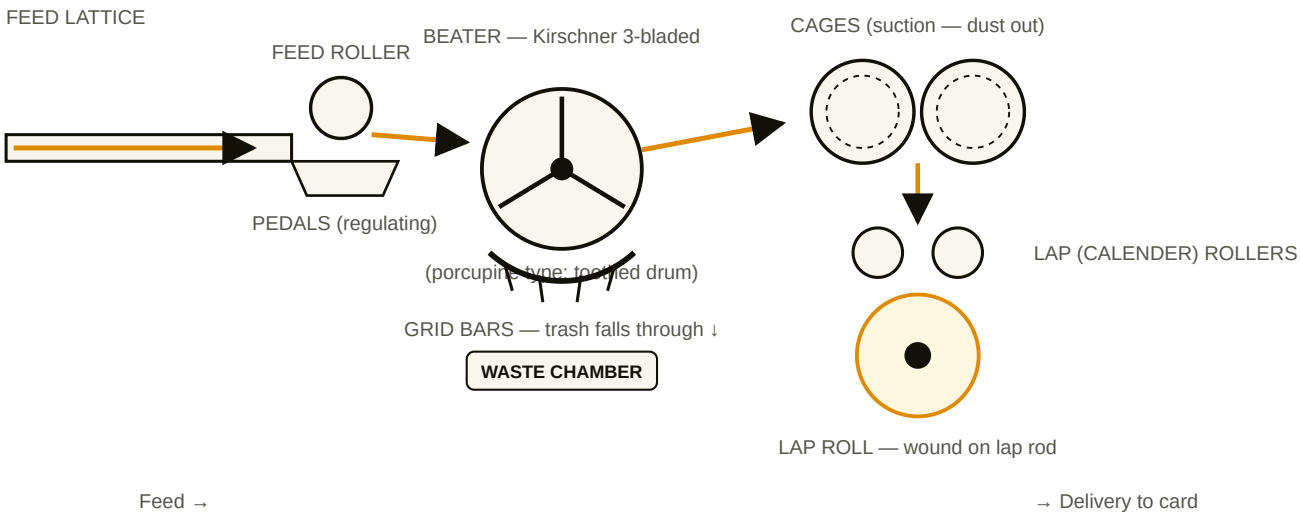


Figure 2 — Scutcher cross-section. Regulated feed presents the sheet to the beater; trash ejects through the grid bars, dust is drawn through the cages, and the cleaned sheet is compressed and wound into a lap.

4. Cleaning intensity and its control

Cleaning intensity — how hard the line works the fibre — must match the cotton: too little leaves trash for the card; too much breaks fibres, creates neps and throws good fibre into waste. Three working controls (air settings — conveying strength and cage suction — are the fourth):

- **Beater speed.** Higher speed (e.g. a scutcher beater raised within its maker's range, about 800 toward 1,000–1,100 rpm) gives more and harder strikes: cleaning improves, but fibre rupture, neps and lint loss rise. Raised for trashy cotton, lowered for long-staple cotton.
- **Grid-bar setting.** The gap between grid and beater and the openness/angle of the grid decide how much material may fall. A more open grid removes more trash *and* more good fibre — waste % rises with cleaning; a closed grid keeps fibre but passes trash to the card.
- **Feed rate.** At fixed beater speed, faster feed means fewer strikes per kilogram and a thicker fringe in which trash stays locked: cleaning per kilogram falls. Slower feed cleans better at the cost of production. Together, speed, grid and feed set the true intensity — strikes per gram of fibre.

CHANGE	CLEANING	WASTE %	RISK IF OVERDONE
Beater speed ↑	Improves	Rises	Fibre breakage, neps, lint in waste ↑
Grid more open	Improves	Rises markedly	Good fibre lost with the trash
Feed rate ↓	Improves per kg	Rises slightly	Production falls; cards starved
Feed rate ↑	Falls per kg	Falls (trash retained)	Trash passes to card; card load ↑

5. Blow-room waste — trash, lint and efficiency

$$\text{Waste \%} = (\text{waste mass} \div \text{feed mass}) \times 100$$

Waste % alone says nothing about *what* was removed. Blow-room waste has two components: **trash** (leaf, seed, husk, sand, dust — its removal is success) and **lint** (spinnable fibre carried out with the waste — pure loss, since the mill bought it as cotton). Some lint loss is unavoidable, but an over-open grid can throw away more value as lint than it saves in trash. Cleaning quality is therefore measured by **cleaning efficiency**:

$$\text{Cleaning efficiency \%} = (\text{trash removed} \div \text{trash in feed}) \times 100$$

Waste samples are separated into lint and trash fractions in the laboratory (trash analyser). The complete verdict on any setting is the trio: waste %, cleaning efficiency, and lint content of the waste. Chasing waste % alone, in either direction, is a classic beginner's error.

WORKED EXAMPLE — LAP PRODUCTION AT THE SCUTCHER

A finisher scutcher delivers lap of 0.400 kg/m at 12 m/min. Production per minute = 0.400 kg/m × 12 m/min = 4.80 kg/min.

Production per hour = 4.80 × 60 = **288 kg/h**. In one line: 0.400 × 12 × 60 = 288 kg/h. ✓ At 90 % efficiency, useful production = 288 × 0.90 = 259.2 kg/h.

WORKED EXAMPLE — CLEANING EFFICIENCY AND LINT IN WASTE

Feed 1,000 kg at 4 % trash; waste removed 60 kg containing 30 kg trash. Trash in feed = 1,000 × 4 ÷ 100 = **40 kg**.

(a) Waste % = 60 ÷ 1,000 × 100 = **6 %**. (b) Cleaning efficiency = 30 ÷ 40 = 0.75 = **75 %**; trash passing to the card = 40 − 30 = 10 kg.

(c) Lint in waste = 60 − 30 = **30 kg** — half the waste is good fibre: the setting cleans reasonably, but expensively.

6. The lap — weight, length, evenness and defects

Weight and length. Lap weight is specified as mass per unit length (g/m; older mills used oz/yd) to suit the card, and checked by weighing a measured cut length every shift and at every mixing change. Total lap weight = weight per metre × length: a 0.400 kg/m lap of 40 m weighs 0.400 × 40 = 16.0 kg. Lap length, set by the knock-off motion, must be constant because card loading assumes it. **Evenness** is judged lengthwise (metre-to-metre weight variation, controlled by the pedal regulation and steady hopper feed; a good finisher lap holds roughly 1–2 % CV between metre lengths) and crosswise (edge-to-edge distribution, controlled by hopper and condenser distribution and undamaged cages).

DEFECT	APPEARANCE	PRINCIPAL CAUSES
Thick lap	Weight above target over long lengths	Feed set too high; pedal regulation faulty; hopper overfilled
Thin / patchy lap	Light places, holes, clouds	Starved or uneven feed; condenser air leaks; damaged lattice
Lap licking	Layers stick and tear on unwinding at the card	Excess lap-roller pressure; damp or oily fibre; sticky (honeydew) cotton; lap wound too tightly
Split / ragged edges; conical lap	Frayed edges, lengthwise split, one side larger	Uneven crosswise feed; damaged cage ends; bent lap rod; rollers not parallel

Lap licking deserves emphasis: its damage appears one machine later, when tearing layers feed surges of thick and thin material into the card — sliver variation often blamed on the card whose true cause is scutcher pressure or damp fibre. The lap is the first uniform product of the mill; its faults set the ceiling for everything after it.

Glossary

Blow room — the line of opening, cleaning and blending machines ending at the scutcher. **Scutcher** — the final blow-room machine: final beating, feed regulation, lap formation. **Lap** — the rolled fibre sheet

of set weight per metre fed to the card. **Condenser** — a suction perforated drum separating fibre from conveying air. **Grid bars** — the adjustable grid beneath a beater through which trash falls. **Kirschner beater** — 3-bladed pinned beater, relatively gentle. **Porcupine beater** — toothed-drum beater, aggressive opening. **Cages** — the scutcher's suction drums condensing the sheet and extracting dust. **Pedal feed** — thickness-sensing regulation of feed-roller speed. **Cleaning efficiency** — trash removed as % of trash in feed. **Lint** — spinnable fibre; lint in waste is fibre lost with trash. **Lap licking** — sticking of lap layers, tearing on unwinding.

Key points

• Objects: gradual opening, cleaning, mixing/blending, uniform lap formation — all four together.
• Line logic: gentle first, intensive later; every cleaning point pairs an opening action with a drop opening.
• Scutcher = pedal-regulated feed + beater over grid bars + suction cages + lap rollers; Kirschner is gentler, porcupine more aggressive.
• Intensity is set by beater speed, grid setting and feed rate — and judged by waste %, cleaning efficiency and lint in waste together.
• Lap quality = correct weight per metre and length, even lengthwise and crosswise; thick lap and lap licking start at the scutcher but damage yarn via the card.
• Production = lap weight (kg/m) × delivery (m/min) × 60; efficiency = trash removed ÷ trash in feed × 100.

Practice

1. A scutcher delivers lap at 0.450 kg/m and 10 m/min. Find the theoretical hourly production and the useful production at 88 % efficiency.
2. Feed of 2,000 kg contains 5 % trash. Waste is 140 kg containing 85 kg trash. Find waste %, cleaning efficiency, trash passed to the card, and lint in waste.
3. Put in line order and state the function of each: condenser, scutcher, bale plucker, hopper feeder, step cleaner.
4. Waste rises from 5 % to 8 % after the grids are opened, but yarn quality does not improve. Using the trash–lint distinction, explain what has probably happened and how you would verify it.
5. Describe the passage of fibre through the scutcher, naming the part responsible for (a) feed regulation, (b) final cleaning, (c) dust removal.
6. Laps from one scutcher tear on unwinding and damage card sliver evenness. Name the defect, give three causes and a remedy for each.

Answer key

1. Per minute: 0.450 × 10 = 4.50 kg/min. Per hour: 4.50 × 60 = **270 kg/h**; at 88 %: 270 × 0.88 = **237.6 kg/h**.
2. Trash in feed = 2,000 × 0.05 = 100 kg. Waste % = 140 ÷ 2,000 × 100 = **7 %**. Efficiency = 85 ÷ 100 = **85 %**. Trash to card = 100 − 85 = **15 kg**. Lint in waste = 140 − 85 = **55 kg**.
3. Bale plucker (small tufts from every bale — first opening, mixing) → step cleaner (grids remove coarse trash) → hopper feeder (reservoir, metered feed, further mixing) → condenser (air separation on suction cage, dust out, sheet formed) → scutcher (final cleaning, regulated feed, lap formation).
4. The open grid is ejecting good fibre with the trash: waste % rose through lint, not through extra trash removed, so cleaning efficiency is unchanged. Verify by trash-analyser separation of waste samples before and after the change, comparing trash removed and lint fraction at equal feed.
5. Fibre rides the feed lattice under the feed roller against the pedals — (a) pedal regulating motion varies feed speed with sheet thickness; the fringe is struck by the beater over the grid bars — (b) beater-and-grid action gives final cleaning; air carries fibre to the suction cages — (c) cages extract dust — and the sheet is compressed by lap rollers and wound onto the lap rod.
6. **Lap licking.** Excess lap-roller pressure → reduce to specification; damp / unconditioned fibre → condition bales and hold humidity in range; lap wound too tightly → check calender pressure and winding setting; sticky cotton → blend down the sticky stock.
